

# FORKAUDIT: A RECEIPT-BOUND OWNERSHIP AUDIT FOR A SEQUENTIAL HYBRID LLM STACK

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## ABSTRACT

Forking many continuations from one long prefix raises two separate questions: whether request state is isolated and which memory denominator is saved. In a hybrid model, token-indexed attention KV and recurrent state make those claims easy to conflate. We present FORKAUDIT, a receipt-bound audit contract, and apply it to one frozen Qwen3.5-35B-A3B configuration. We distinguish three evidence levels: (i) receipt-complete lifecycle/storage predicates conditional on honest producer capture and external byte bindings; (ii) selected independent operator corroboration; and (iii) positive-control gate reachability. Across eight pre-specified books, 96 configurations cross full-copy/shared-document KV with materialized/borrowed GDN bases at  $N \in \{1, 8, 32\}$ . Registered token, canonical full-logit, logical-KV, and final-GDN digests match across all four cells and 288 adjacent-fan-out comparisons; phase receipts show post-transition mutable GDN state disjoint from bases and peers. Separately, eight attention rows pass a dense-FP32 oracle ( $L_2 \leq 0.001743 < 0.005$ ), four preregistered GDN transitions pass a NumPy recurrence, and nine live faults reach intended gates. Per-call compiled-binary/autotuning provenance remains unbound. At  $N = 32$ , shared KV reduces median final allocated delta from 4.90 to 2.23 GiB. Unpooled Mac, H20, HYPIC, and serving controls provide context, not audit evidence. The result is evidence about a captured receipt ledger for one sequential stack—not independent producer recapture, end-to-end verification, or production capacity.

## 1 INTRODUCTION

Best-of- $N$  decoding, agents, tree search, and document question answering create many continuations from a common prefix. Modern serving systems exploit this structure through paged KV caches, prefix trees, and scheduling policies (Kwon et al., 2023; Zheng et al., 2024; Gim et al., 2024; Srivatsa et al., 2025). For a Transformer, “sharing the prefix” can often be described as sharing read-only KV blocks and giving each continuation private append blocks. Hybrid models complicate this picture: full-attention layers coexist with recurrent or linear-attention layers whose compact state is updated in place unless the runtime installs an explicit request-local view (Yang et al., 2025; Lenz et al., 2025; Pan et al., 2025).

This creates two evaluation hazards. First, matching output tokens does not prove request isolation: two branches can agree while aliasing the same mutable state, and a bug can affect both compared implementations. Second, “memory saving” may refer to logical payload, allocated cache pages, framework allocator peaks, process-level device memory, or admitted service capacity. Those quantities are not interchangeable. An impressive cache-pool ratio may be small after model weights and common runtime state enter the denominator. An archival Mac control makes the first hazard concrete: two quantized CoMem variants agree exactly with their shared BF16 split reference while the whole split family disagrees substantially with vanilla dense (Table 1).

We propose FORKAUDIT, a correctness-first receipt contract for a captured ledger under trusted producer capture. It partitions document and request state, records page reservations and phase-aware recurrent-state rebinding, fingerprints the Python attention callable, and combines a  $2 \times 2$  ownership factorial with relational and independent numerical checks. A mutation specification and gate assignment fixed before candidate execution then ask whether each advertised gate is sensitive

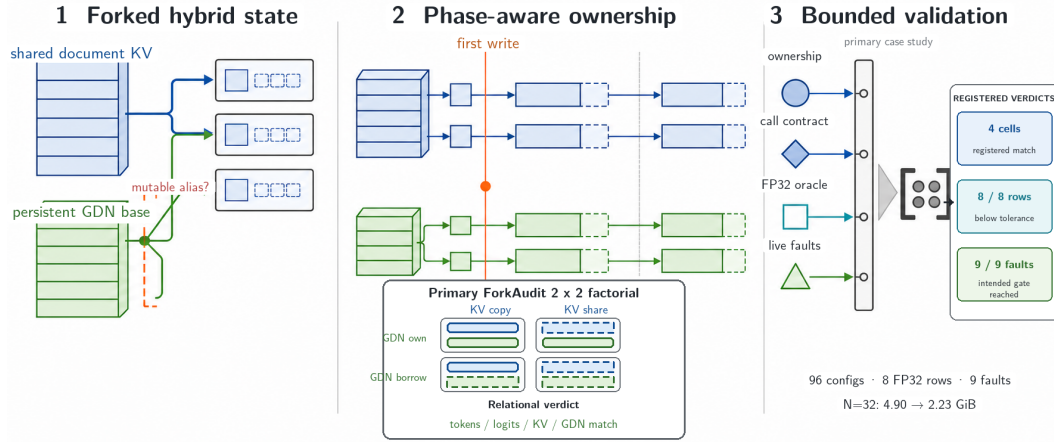


Figure 1: FORKAUDIT in one view. A common prefix forks into requests whose token-indexed KV and recurrent GDN state obey different lifetime rules. The contract crosses four setup policies, requires private append pages and post-write recurrent rebinding, and exposes three non-equivalent levels: trusted-capture receipt completeness, selected independent operator checks, and positive-control gate reachability. The primary audit evidence shown here comprises 96 factor configurations, eight checked attention rows, nine targeted faults, and the  $N = 32$  allocation endpoint; per-call compiled dispatch remains open. These are bounded one-model, one-schedule results, not speedup or independent recapture claims. This explanatory schematic is not experimental evidence; the registered tables and artifacts authorize the numerical claims.

to a corresponding live fault. The measurement side reports page geometry separately from allocator deltas and phase-bounded peaks. FORKAUDIT is not a caching policy, kernel, or scheduler; its purpose is to make the captured-receipt boundary of one frozen sequential implementation explicit and machine-checkable; it does not independently attest to the producer, and its per-call compiled dispatch remains open.

The primary contribution is the receipt-replayable audit contract; the frozen sequential case and accounting reconstruction below are validation artifacts for that contract, not separate mechanisms. Concretely, we provide:

- **Receipt-complete lifecycle/storage targets under trusted capture.** The contract maps immutable document KV, private append pages, borrowed versus materialized GDN bases, first-transition rebinding, tail COW, and callable provenance to typed records. Lifecycle/storage predicates replay completely; compiled dispatch is partial (Sections 3–4).
- **A relational factorial plus selected independent checks.** Four KV-by-GDN ownership cells have identical registered token sequences and shape/dtype-bound CPU-FP32 logit, logical-KV, and GDN byte digests across eight pre-specified windows and three fan-outs. This equality is relational and cannot exclude a common-mode defect; separately, selected full-attention rows pass an IEEE-FP32 dense oracle, and selected recurrent-core transitions pass a separate NumPy recurrence oracle, under tolerances fixed before execution (Section 5).
- **Positive-control reachability and explicit denominators.** Nine live mutants reach their intended gates under matched-clean controls, without estimating detection rate or completeness, while allocator endpoints show which savings survive once model and runtime state enter the denominator. A separate same-adapter cohort further exercises aligned-prefix cancellation, zero-scrub, exact reclamation, and stale-lease rejection without changing outputs or final state.

Relative to ordinary pass/fail tests, FORKAUDIT adds a joint replay obligation: frozen roles, setup–first-transition–final lifecycle, storage/call/accounting records, and a predeclared fault map must close together. The constituent checks are not individually new or exhaustive.

The supported claim is deliberately narrow: conditional receipt-level ownership evidence for one frozen sequential stack. The primary audit uses one model, one hardware family, 16-bit KV, one 4,095-token partial-tail factorial and one aligned 4,096-token cancellation lifecycle, batch size one per request, and sequential round-major model steps on one CUDA stream. The independent oracles cover selected attention and recurrent-core operators conditional on captured internal inputs, not the full hybrid model. A separate single-request deployment cohort reports TTFT, TPOT, throughput, reusable-state bytes, and F1 for vanilla dense, full-prefix cache reuse, and CoMem quantization; it does not measure continuous batching, scheduler throughput, NVML capacity, or maximum admitted concurrency. The nearest executable hybrid-cache baseline uses the authors’ HYPIC code on the same checkpoint, slice, H20-3e hardware, input cap, and decoding budget; its Full Recompute, Prefix Cache, and full HYPIC arms use TP=1 and one streaming client. The published Qwen3.5 panels use TP=2, so this is a protocol-matched adaptation rather than a reproduction of their published latency. An independent receipt-instrumented cohort measures retained-document tensor payload for Prefix Cache and HYPIC in that same fixed configuration (eight workloads per arm). A separate serving-framework panel tests vLLM and SGLang cache-off/on controls; each experimental block is interpreted on its own.

## 2 BACKGROUND AND RELATED WORK

**Paged and prefix-aware inference.** PagedAttention, Prompt Cache, RadixAttention, ChunkAttention, Hydragen, and Preble establish block sharing, prefix reuse, shared-prefix computation, and scheduling as prior art (Kwon et al., 2023; Gim et al., 2024; Zheng et al., 2024; Ye et al., 2024; Juravsky et al., 2024; Srivatsa et al., 2025). Our narrower question is what execution evidence demonstrates isolation when several state types coexist.

**Hybrid state and cache management.** Gated Delta Networks maintain mutable compact state unlike token-indexed KV (Yang et al., 2025); Jamba is a hybrid architecture and Marconi manages hybrid-model prefixes (Lenz et al., 2025; Pan et al., 2025). HYPIC caches segment transition operators plus zero-start end states and recomputes a small seam to enable position-independent reuse in hybrid stacks (Liu et al., 2026c). Palu, SqueezeAttention, and HeadKV instead study KV compression or budgeting (Chang et al., 2025; Wang et al., 2025; Fu et al., 2025). FORKAUDIT does not replace these policies: it audits their ownership boundary. Appendix Table 11 keeps their original-protocol numbers separate from our common-model controls.

**Persistent intermediate residual reuse.** CoMem makes decoder split depth  $j$  a reusable-context axis (Liu et al., 2026a;b). Its WRITE stage persists a depth- $j$  residual per document token, SELECT retrieves bounded chunks, and READ executes the query and suffix layers. This motivates our persistent document object. A bounded depth-7 deployment slice places Q8/Q4/mixed state beside vanilla dense and full-prefix reuse, but FORKAUDIT claims neither a new selector nor a new quality-latency frontier.

**Evaluation and metamorphic testing.** SCBench emphasizes cache lifecycles (Li et al., 2025); metamorphic testing, MT-DLComp, and CRADLE motivate preserved relations and independent checks (Chen et al., 2018; Xiao et al., 2022; Pham et al., 2019); DNNMem motivates separate memory denominators (Gao et al., 2020). We combine these established ideas into a heterogeneous-state audit: adding resident branches must preserve registered observables, selected captured-input operators receive independent FP32 checks, and page-pool and allocator quantities remain distinct. Appendix Table 9 gives the component-by-component novelty boundary.

## 3 THE FORKAUDIT CONTRACT

Consider a document prefix  $D$  and resident requests  $r \in \{1, \dots, N\}$ . Let  $\mathcal{A}$  be the full-attention layers and  $\mathcal{G}$  the recurrent linear-attention layers. For  $\ell \in \mathcal{A}$ , the state is a read-only set of document pages  $K_\ell^D$  plus a request-private reservation  $K_\ell^r$ . For  $\ell \in \mathcal{G}$ , it is a persistent prefix base  $G_\ell^D$  plus the request-local recurrent state  $G_\ell^r$ . A logical fork is

$$\mathcal{F}_r(D) = (\{K_\ell^D, K_\ell^r\}_{\ell \in \mathcal{A}}, \{G_\ell^D, G_\ell^r\}_{\ell \in \mathcal{G}}). \quad (1)$$

The checklist has seven audit targets. A target may be fully witnessed, partially witnessed, or left open; passing the case study must not silently upgrade a partial witness into a proof.

1. **Frozen identity.** Model, data, query bank, layer geometry, kernel descriptor, and protocol hashes are fixed before execution.
2. **Prefix immutability.**  $K_\ell^D$  and  $G_\ell^D$  have identical digests before and after all continuations.
3. **Private ownership.** KV reservations of distinct shared-document requests are disjoint in the common arena; independent materialized arenas have disjoint storage. At setup, a GDN request either owns a materialized base or holds an explicitly read-only persistent-base alias. After its first transition, all 60 mutable tensors must occupy ranges disjoint from the base and every peer.
4. **Tail-safe append.** If the last document page is partial, its valid tokens are copied once to a private page before a request appends.
5. **Dispatch provenance.** Both controls invoke the same recorded attention Python callable with the same query, mask, position, and append contracts; fallback and full-KV concatenation are forbidden. This does not fingerprint a compiled Triton binary or autotuning choice.
6. **Cross-arm equivalence.** For the same query and schedule, the materialized and shared ownership arms must have exactly equal generated tokens, full-vocabulary logits, final logical KV, and request recurrent state.
7. **Cross- $N$  prefix consistency.** For each query observed at two or more fan-outs, its result at every larger fan-out must match its smallest- $N$  result.

**Minimal adoption workflow and decision rule (normative).** Applying FORKAUDIT to a sequential hybrid runtime requires four steps. (1) *Freeze and assign roles*: bind code, model, data, protocol, and the mapping from runtime objects to document, base, request, and private-reservation roles before candidate outputs. (2) *Capture mandatory records*: emit setup, first-transition, and final lifecycle events together with canonical content digests, normalized storage ranges, ordered append/dispatch receipts, semantic outputs, and named allocator snapshots. (3) *Replay the seven targets blindly*: an independent aggregate recomputes every applicable predicate from frozen records; a target is receipt-complete (RC) only when all mandatory records exist, *partial* when evidence is sampled or proxy-level, and *open* when a mandatory record is absent or replay fails. (4) *Add fault records only for fault-sensitivity claims*: freeze the target-to-gate map before outputs and retain clean false positives, wrong-gate detections, and escapes as valid negatives. RC remains conditional on honest producer capture and external byte bindings; it is not independent producer recapture or end-to-end or exhaustive correctness. In the primary run, terminal identity is RC but per-call compiled dispatch and autotuning provenance remain partial. We evaluate this workflow on the fixed Qwen runtime below. Appendix D gives the exhaustive target-to-record dependency matrix and the pointer-free storage schema used by this audit.

Appendix Table 15 gives the four normative record classes and their replay obligations.

Here and below, “exact” for a registered tensor-valued observable means equality of a canonical serialization: contiguous CPU-FP32 bytes with shape and dtype bound into the record, or the correspondingly specified logical-KV or GDN byte digest. It is not a claim that an unarchived device tensor was compared bit-for-bit before canonicalization.

Targets 6 and 7 are relational self-consistency oracles. They can reveal a layout-specific error or an  $N$ -dependent change in recorded observables, respectively. The primary factorial additionally captures the same post-RoPE query, logical K/V, position values, mask, and effective scale for one rank-specific full-attention row selected by the pre-execution static plan and compares the fused result with dense IEEE-FP32 attention. A separate recurrent oracle captures four transitions originally selected at layers 0, 10, 20, and 38 after the native BF16 q/k normalizer, then evaluates the transition with a candidate-import-free NumPy FP32 token recurrence. A fresh single preregistration, frozen before the successful run, binds this post-native-normalization boundary, the implicit query scale  $1/\sqrt{128}$ , the four rows, tolerances, and seeded faults. These are independent numerical checks for selected operator boundaries, not for upstream activation construction, q/k normalization itself, every GDN layer, or the complete end-to-end model.

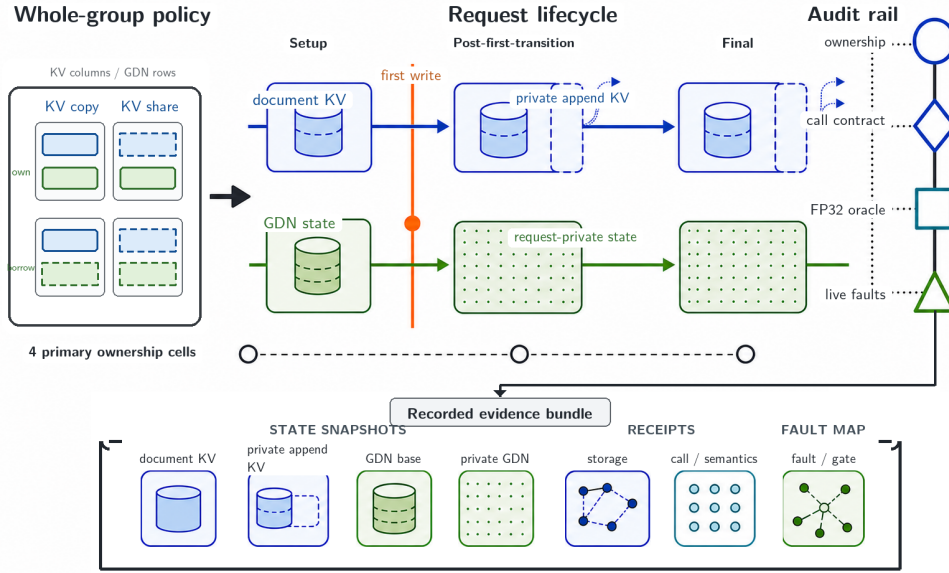


Figure 2: ForkAudit’s ownership-factorial architecture. The two left-hand axes select setup ownership for the entire request group; they never assign different policies to individual requests. KV document ownership remains policy-dependent while append tails are private. A borrowed GDN base may alias the persistent base only at setup; after a request’s first transition, its mutable GDN state is private and disjoint from the base and peers. The audit spine denotes replay classes, not additional execution stages or numerical results. The bracketed bottom row is the evidence bundle recorded by that audit. From left to right it contains the document-KV snapshot, private-append snapshot, persistent-GDN base, request-private GDN state, storage-binding graph, call/semantic receipt, and fault-to-gate map. Offline replay validates this bundle. Within the trusted-capture boundary, lifecycle/storage targets are complete, dispatch is partial, and numerical/fault checks are selected support. The bitmap is an explanatory schematic, not experimental evidence; its technical labels and relations were deterministically overlaid and checked against the contract.

Figure 2 shows the four primary-factorial cells and the lifecycle they share. The KV axis contrasts one document-plus-private arena per request with a shared document and disjoint private tail/append pages. The GDN axis contrasts per-request setup materialization with a borrowed immutable base followed by the same functional rebind. Memory and witness cells are rebuilt separately so storage guards do not retain tensors during allocator measurement.

## 4 RECEIPT-BOUND AUDIT PROTOCOL

### 4.1 FROZEN CASE-STUDY GEOMETRY

We instantiate and validate the contract in one fixed runtime configuration: Qwen3.5-35B-A3B (Qwen Team, 2026a;b): ten vLLM-0.26 Triton full-attention layers with BF16 (artifact “Q16”) KV, 128-token pages, and thirty Transformers-torch GDN layers, run sequentially on eight H20-3e GPUs. The ledger binds the Python attention callable but not the compiled binary; each request’s GDN digest covers 60 tensors. Every ForkAudit correctness claim is conditional on this stack.

Eight ranks use distinct PG-19 training books (Rae et al., 2020), each with one 4,095-token document and 32 non-overlapping queries. The four-cell factorial evaluates  $N \in \{1, 8, 32\}$  and eight argmax steps; 39 query/generated tokens enter KV, so the 127-token final document page exercises tail COW. Appendix Table 4 separates this run from all context cohorts.

## 4.2 AUDIT SEQUENCE

Each cell hashes its frozen identity, builds document KV and persistent GDN state, separately rebuilds allocator and witness cells, records setup—first transition—final ownership, and executes one query plus seven continuation steps. It then compares registered tokens, logits, logical KV, and GDN state; replays fixed-axis projections and one dense-FP32 sidecar per rank; and runs matched-clean/live-fault pairs from fresh state. Aggregation requires every raw-byte binding. Appendix Table 5 states target strength; the request—base predicate is directly exercised at  $N = 1$ .

## 4.3 CONTROLS, SCHEDULE, AND ORACLES

All requests are resident before sequential round-major steps on one CUDA stream; this is not concurrent-kernel evidence. Full logits are canonicalized as shape/dtype-bound contiguous CPU-FP32 bytes, while logical KV, 60 GDN tensors, immutable bases, call positions/masks/scales, storage, and allocator records receive typed digests. Replay closes these records, factor projections, oracle math, and fault receipts, but cannot reconstruct unarchived device tensors or independently recapture the producer.

The GDN oracle was preregistered before its successful execution with one window, four layers, a post-native-q/k-normalization boundary, scale  $1/\sqrt{128}$ , tolerances, and four wrong recurrences. A prior discarded diagnostic informed that boundary; it is chronology, not evidence from the successful run.

## 4.4 MEMORY DENOMINATORS

For the primary ForkAudit experiment, we keep logical state, allocated pages, synchronized PyTorch current/peak allocated/reserved bytes, and unmeasured process/service capacity distinct; only pages and allocator bytes support its quantitative memory claims, and phase-reset peaks are labeled by their measurement window. The deployment comparison uses a separate Store estimand: the union of physical tensor byte ranges owned by the retained document entry at its measurement boundary. It excludes Python metadata, preallocated pools, NVML deltas, and admission capacity.

## 4.5 GOVERNANCE AND LIVE MUTATION CHECKS

Pre-output ledgers bind code, model, data, queries, and protocol; the formal primary rerun also verifies matching terminal source/model identity. Nine live mutations cover allocation/binding, tail COW, GDN aliasing, positions, masks, callable identity, and paged-view type; each has a fresh matched control and cleanup receipt. These tests establish intended-gate sensitivity, not completeness.

# 5 RESULTS

## 5.1 UNPOOLED MAC MOTIVATION CONTEXT EXPOSES COMMON-MODE AGREEMENT

Before the H20 audit, an archival Apple-M4-Pro control provides a compact counterexample to treating implementation agreement as an independent oracle. On the same six-document/six-query selected packs, vanilla dense MLX is the external reference, while CoMem (Liu et al., 2026a;b) reuses a depth-7 split residual in BF16, Q8, or Q4. Across five fresh processes, both quantized variants retain every answer-position top-1 decision of the BF16 split path, yet that entire split family agrees with vanilla dense on only 71.89% of answer positions. Table 1 therefore motivates the later independent attention/GDN oracles; it is a small teacher-forced mechanism control, not primary ForkAudit evidence, a second ForkAudit deployment, or a cross-hardware performance comparison.

## 5.2 UNPOOLED SAME-CHECKPOINT DEPLOYMENT AND NEAREST-WORK CONTEXT

Table 2 combines two separate conventional end-to-end timing/quality blocks with an independent retained-state receipt cohort; no ForkAudit ownership conclusion depends on them. All use the same Qwen3.5 checkpoint, the same eight LongBench validation items, one H20-3e per item, a 4,096-token cap, greedy decoding, and at most 32 generated tokens. The first timing block uses our in-process

Table 1: Unpooled Mac motivation context (Apple M4 Pro, five fresh MLX processes; median over 30 query executions at split depth 7). Vanilla dense is the external reference. CoMem BF16/Q8/Q4 variants use the same selected documents. Store size is the six-document residual corpus; online time includes query Write, dequantization, and suffix Read. This small teacher-forced demonstration set is not primary ForkAudit evidence and supports no cross-hardware or production-speed claim.

| Method               | Store (MiB) | Online (ms) | Speedup | Top-1 vs. dense |
|----------------------|-------------|-------------|---------|-----------------|
| Vanilla dense prompt | –           | 194.70      | 1.00×   | 100.00%         |
| CoMem BF16 split     | 1.998       | 168.88      | 1.15×   | 71.89%          |
| CoMem Q8 split       | 1.061       | 169.17      | 1.15×   | 71.89%          |
| CoMem Q4 split       | 0.562       | 169.25      | 1.15×   | 71.89%          |

Q8/Q4 retain 100% top-1 agreement with the BF16 split path, yet all split rows inherit only 71.89% agreement with vanilla dense. The speed column is a small teacher-forced mechanism benchmark, not production RAG throughput.

Transformers deployment harness for vanilla dense, a full-prefix KV reuse baseline, and CoMem’s Q8/Q4/mixed state variants. The second uses the HYPIC authors’ official commit on SGLang 0.5.14. Because the runtimes and timing harnesses differ, latency and throughput are compared within blocks; we do not compute a cross-block speedup.

The rows expose tradeoffs, not a universal win. Full-prefix reuse is fastest (13.38 tokens/s, 0.163 s TTFT) but retains 140.34 MiB/document. CoMem Q8 keeps 39.14 mean F1 with 15.89 MiB at 7.15 tokens/s; mixed Q4/Q8 keeps 39.01 F1 with 10.01 MiB, while all-Q4 reaches 36.09 F1 with 8.41 MiB. These eight workloads are context, not a complete LongBench or robustness result.

HYPIC (Liu et al., 2026c), the closest executable hybrid-cache baseline, uses the released Full Recompute, Prefix Cache, and `transition_rope_recompute` arms with its eight-token seam. In this TP=1 block, Full Recompute gives 39.14 F1/0.217 s TTFT/14.77 tokens/s; Prefix Cache gives 39.14/0.072/19.07; HYPIC gives 39.63, 0.101 s, and 17.67. Published Qwen3.5 panels use TP=2, so our rows are a protocol-matched adaptation, not a latency/QPS reproduction. Separately, a receipt-instrumented 8+8-cell cohort measures the retained-document tensor payload after the first request boundary for Prefix Cache and HYPIC under the same fixed model, workloads, and TP=1 configuration.

Table 2: Unpooled same-checkpoint H20 context on the eight-item Qasper/2WikiMQA slice. The first block is an in-process Transformers deployment cohort with three timing repeats per item. HYPIC (Liu et al., 2026c) timing/F1 comes from a separate 24-cell TP=1 official-code cohort; its Store values come from an independent receipt-instrumented 8+8-cell cohort. All use Qwen3.5-35B-A3B, one H20-3e per item, a 4,096-token cap, greedy decoding, and at most 32 generated tokens. Timing is interpreted within block; no cross-block speedup, production-capacity, broad-quality, or ForkAudit-ownership claim is made.

| Method                                                                                 | Persistent state   | Store (MiB) | TTFT (s) | TPOT (ms) | tok/s | F1    |
|----------------------------------------------------------------------------------------|--------------------|-------------|----------|-----------|-------|-------|
| <i>Transformers 5.14.1 in-process deployment cohort</i>                                |                    |             |          |           |       |       |
| Vanilla dense                                                                          | –                  | –           | 0.649    | 648.75    | 1.36  | 39.42 |
| Full-prefix KV cache                                                                   | Q16                | 140.34      | 0.163    | 54.11     | 13.38 | 39.14 |
| CoMem state                                                                            | Q8                 | 15.89       | 0.673    | 55.44     | 7.15  | 39.14 |
| CoMem mixed state                                                                      | Q4/Q4/Q8           | 10.01       | 0.674    | 54.97     | 7.19  | 39.01 |
| CoMem per-layer mixed                                                                  | mixed              | 9.74        | 0.673    | 55.07     | 7.19  | 39.11 |
| CoMem state                                                                            | Q4                 | 8.41        | 0.674    | 55.02     | 7.19  | 36.09 |
| <i>HYPIC commit 98147c0 timing/F1; receipt-instrumented Store; SGLang 0.5.14, TP=1</i> |                    |             |          |           |       |       |
| Official HYPIC code: full recompute                                                    | –                  | –           | 0.217    | 49.21     | 14.77 | 39.14 |
| Official HYPIC code: prefix cache                                                      | Radix prefix       | 139.53      | 0.072    | 50.15     | 19.07 | 39.14 |
| HYPIC (transition + seam 8)                                                            | segment transition | 324.09      | 0.101    | 52.13     | 17.67 | 39.63 |

Store is the median retained-document tensor payload: persistent document bytes for CoMem/full-prefix reuse and the union of target-entry-owned physical tensor byte ranges for Prefix Cache/HYPIC. It excludes Python metadata, preallocated pools, NVML/process deltas, and admission capacity. HYPIC timing/F1 uses streaming-client wall-clock from 24 cells; HYPIC Store uses a separate 16-cell externally replayed receipt cohort. Published Qwen3.5 HYPIC panels use TP=2, whereas this fixed same-slice adaptation uses TP=1.

Across the independent Store cells, Prefix Cache retains a median 139.53 MiB and HYPIC retains 324.09 MiB under the same target-owned physical-tensor-range denominator, so HYPIC retains 2.32× the Prefix Cache payload. CoMem Q8’s 15.89 MiB is 8.78× smaller than Prefix Cache

and  $20.39\times$  smaller than HYPIC under that Store estimand. This comparison excludes metadata, preallocated pools, and NVML/process deltas; it is a retained-state result, not a capacity or cross-runtime speedup claim. HYPIC remains faster within its own timing cohort, while CoMem supplies the smaller explicit state–quality surface.

**Further unpooled context.** vLLM and SGLang each record 8/8 prefix hits and preserve 8/8 cache-off predictions; timings remain within framework (Table 13). Hydragen matches one dense layer-3 family within relative  $\ell_2 < 0.0027$  at  $N = 8/32$ , while Palu whitening improves held-out K/V projection error over SVD at ranks 64/128/192. Table 12 treats these as bounded operator compatibility, not a leaderboard, all-layer result, or combined system.

*Primary result status and evidence hierarchy.* In the seven-target workflow, frozen identity, prefix immutability, private ownership, tail-safe append, cross-arm equivalence, and cross- $N$  consistency (targets 1–4 and 6–7) are RC only within trusted capture; dispatch provenance (target 5) is partial because no per-call compiled-binary or explicit autotuning-choice binding is available. Selected oracles provide numerical corroboration, mutations provide gate reachability, and all other tables are unpooled context. Independent producer recapture and end-to-end correctness remain outside the claim.

### 5.3 RELATIONAL EQUALITY HOLDS FOR EVERY REGISTERED OBSERVABLE

The primary factorial executes  $8 \times 3 \times 4 = 96$  rank–fan-out–factor configurations. Each configuration is rebuilt once for allocator measurement and once for ownership/semantic witnesses, for 192 live cells total. The blind aggregate reports exact token trajectories, full-vocabulary logits, final logical KV, and final GDN values across all four cells. Fixed-axis projections also match: changing only GDN setup ownership leaves the KV lifecycle/source/query/call schedule unchanged, while changing only KV ownership leaves the GDN base values and transition trajectory unchanged. The offline replay additionally reconstructs all 288 adjacent-fan-out comparisons— $N = 1 \rightarrow 8$  and  $8 \rightarrow 32$  for every arm, rank, and shared query identity—with no mismatch. These are relational equalities among the registered cells: they support the receipt-level ownership witness but cannot by themselves exclude a defect shared by every arm.

The range witnesses explain where that equality comes from. A borrowed GDN cell begins with exactly 60 immutable persistent-base aliases per request; a materialized cell begins request–base and request–request disjoint. After a request’s first model transition, both policies bind 60 request-private mutable tensors whose byte ranges are disjoint from the persistent base and all completed peers. Incomplete borrowed requests remain exact read-only aliases until their own transition. These predicates pass at setup, first transition, and post-generation for every factor configuration. In particular, the  $N = 1$  cells directly exercise request–base disjointness rather than relying on a vacuous peer comparison.

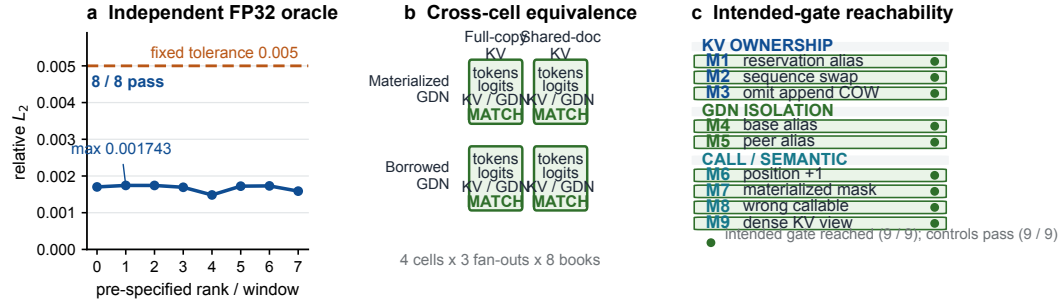


Figure 3: Primary-factorial validation. Left: every pre-specified full-attention row is below the independently pre-fixed 0.005 relative- $L_2$  tolerance. Center: all four KV-by-GDN ownership cells have identically registered canonical observables at all three fan-outs. Right: each live fault reaches its intended gate under a passing matched control; this is intended-gate reachability, not detector completeness or rate. The panels are, respectively, selected numerical evidence, relational equality, and targeted positive controls—not equivalent assurance levels.

#### 5.4 INDEPENDENT ATTENTION ROWS PASS THE PRE-FIXED TOLERANCE

From physical BF16 blocks, the frozen table, ordered append shadows, and live call metadata, the independent dense-FP32 oracle obtains relative- $L_2$  0.001484–0.001743, below the pre-fixed 0.005 tolerance on all eight selected rows. This is conditional on captured post-RoPE inputs, not an independent upstream or end-to-end implementation.

#### 5.5 SELECTED RECURRENT CHECKS AND TARGETED POSITIVE CONTROLS

At layers 0, 10, 20, and 38, the fully preregistered candidate-import-free NumPy recurrence matches clean outputs and states (maximum relative  $L_2$  0.001652 and  $1.40 \times 10^{-7}$ ) and rejects four seeded wrong recurrences. Appendix Tables 20 and 7 state the captured boundary and show all nine intended gates under passing controls. Separate lifecycle and deterministic request-step checks preserve clean outputs/state through cancel–scrub–reclaim/replacement: 16 clean geometry cells match and 48 frozen fault trials reach their expected gates (Tables 17–18). These bounded checks are demonstrated in the fixed setup described above; they do not estimate concurrent-kernel behavior or detection rate.

#### 5.6 OWNERSHIP FACTORS MOVE DIFFERENT MEMORY COMPONENTS

At  $N = 32$ , switching full-copy to shared-document KV reduces the final allocated delta by 2.672 GiB with materialized GDN setup and 2.661 GiB with borrowed setup. Borrowing changes allocation timing, not the shared-KV end state: both GDN policies finish at 2.229 GiB. Appendix Table 16 gives the four endpoints and keeps the earlier  $12.33\times$  pool versus  $1.0398\times$  model-inclusive capacity cohort separate.

### 6 DISCUSSION AND LIMITATIONS

**What the experiment establishes.** Conditional on honest capture and the registered byte bindings, receipts from one frozen stack witness request/base and request/peer relations across ten attention and thirty GDN layers. Cross-cell equality, selected FP32 checks, and positive controls are bounded levels, not closure of every common-mode path.

**What it does not establish.** Positive ForkAudit evidence remains one model/H20 family, 16-bit KV, one sequential partial-tail factorial, and bounded same-adaptor lifecycle paths— not concurrent kernels, continuous batching, lower-precision KV, or binary identity. CoMem deployment, vLLM/SGLang, the 24-cell TP=1 HYPIC timing/quality cohort, and the separate 16-cell HYPIC Store cohort are unpooled context.

**Evidence limitations.** Ranks are workloads, not replicates; most tensors ship as digests; replay assumes honest capture; and neither oracle is end-to-end. Appendix I gives the full boundary.

### 7 REPRODUCIBILITY STATEMENT

Manifest-first CPU replays cover RR2, lifecycle/GDN, serving, scheduler, and operator/policy transfers. RR2 rechecks 536 artifacts, 96 factorial cells, 288 cross- $N$  comparisons, eight attention rows, and nine fault pairs. HYPIC’s 24 timing/quality cells bind a 4,404-entry terminal artifact ledger and reaggregate byte-identically; independently, all 16 retained-state cells have raw, pre-measurement, and terminal receipts and pass the hash-bound external replay. The terminal rerun closes source ledgers, 14 model rehashes, and aggregate replay for that run only, not producer recapture; dispatch and autotuning remain unbound. Appendix J maps every package.

### 8 CONCLUSION

For one frozen Qwen3.5 sequential stack, FORKAUDIT makes lifecycle/storage predicates receipt-complete only within trusted capture. Selected oracles are narrower corroboration, and nine mutations are positive controls rather than a detection-rate estimate; per-call compiled dispatch and independent

recapture remain open. Unpooled context establishes neither end-to-end correctness nor concurrent serving, capacity, or a performance frontier.

## REFERENCES

- Chi-Chih Chang, Wei-Cheng Lin, Chien-Yu Lin, Chong-Yan Chen, Yu-Fang Hu, Pei-Shuo Wang, Ning-Chi Huang, Luis Ceze, Mohamed Abdelfattah, and Kai-Chiang Wu. Palu: KV-cache compression with low-rank projection. In *International Conference on Learning Representations*, volume 2025, pp. 50222–50249, 2025. URL [https://proceedings.iclr.cc/paper\\_files/paper/2025/hash/7da6e0e00702c60607a6ae05c802ef85-Abstract-Conference.html](https://proceedings.iclr.cc/paper_files/paper/2025/hash/7da6e0e00702c60607a6ae05c802ef85-Abstract-Conference.html).
- Tsong Yueh Chen, Fei-Ching Kuo, Huai Liu, Pak-Lok Poon, Dave Towey, T. H. Tse, and Zhi Quan Zhou. Metamorphic testing: A review of challenges and opportunities. *ACM Computing Surveys*, 51(1):4:1–4:27, 2018. doi: 10.1145/3143561. URL <https://doi.org/10.1145/3143561>.
- Yu Fu, Zefan Cai, Abedelkadir Asi, Wayne Xiong, Yue Dong, and Wen Xiao. Not all heads matter: A head-level KV cache compression method with integrated retrieval and reasoning. In *International Conference on Learning Representations*, volume 2025, pp. 99269–99290, 2025. URL [https://proceedings.iclr.cc/paper\\_files/paper/2025/hash/f649556471416b35e60ae0de7c1e3619-Abstract-Conference.html](https://proceedings.iclr.cc/paper_files/paper/2025/hash/f649556471416b35e60ae0de7c1e3619-Abstract-Conference.html).
- Yanjie Gao, Yu Liu, Hongyu Zhang, Zhengxian Li, Yonghao Zhu, Haoxiang Lin, and Mao Yang. Estimating GPU memory consumption of deep learning models. In *Proceedings of the 28th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering*, pp. 1342–1352. Association for Computing Machinery, 2020. doi: 10.1145/3368089.3417050. URL <https://doi.org/10.1145/3368089.3417050>.
- In Gim, Guojun Chen, Seung-seob Lee, Nikhil Sarda, Anurag Khandelwal, and Lin Zhong. Prompt cache: Modular attention reuse for low-latency inference. In *Proceedings of Machine Learning and Systems*, volume 6, pp. 325–338, 2024. URL [https://proceedings.mlsys.org/paper\\_files/paper/2024/hash/a66caal703fe34705a4368c3014c1966-Abstract-Conference.html](https://proceedings.mlsys.org/paper_files/paper/2024/hash/a66caal703fe34705a4368c3014c1966-Abstract-Conference.html).
- Jordan Juravsky, Bradley Brown, Ryan Ehrlich, Daniel Y. Fu, Christopher Ré, and Azalia Mirhoseini. Hydragen: High-throughput LLM inference with shared prefixes. *arXiv preprint arXiv:2402.05099*, 2024. URL <https://arxiv.org/abs/2402.05099>.
- Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles*, pp. 611–626. ACM, 2023. doi: 10.1145/3600006.3613165. URL <https://doi.org/10.1145/3600006.3613165>.
- Barak Lenz, Opher Lieber, Alan Arazi, Amir Bergman, Avshalom Manevich, Barak Peleg, Ben Aviram, Chen Almagor, Clara Fridman, Dan Padnos, Daniel Gissin, Daniel Jannai, Dor Muhlgay, Dor Zimberg, Edden Gerber, Elad Dolev, Eran Krakovsky, Erez Sa, Erez Schwartz, Gal Cohen, Gal Shachaf, Haim Rozenblum, Hofit Bata, Ido Blass, Inbal Magar, Itay Dalmedigos, Jhonathan Osin, Julie Fadlon, Maria Rozman, Matan Danos, Michael Gokhman, Mor Zusman, Naama Gidron, Nir Ratner, Noam Gat, Noam Rozen, Oded Fried, Ohad Leshno, Omer Antverg, Omri Abend, Or Dagan, Orit Cohavi, Raz Alon, Ro’i Belson, Roi Cohen, Rom Gilad, Roman Glozman, Shahar Lev, Shai Shalev-Shwartz, Shaked Meirom, Tal Delbari, Tal Ness, Tomer Asida, Tom Ben Gal, Tom Braude, Uriya Pumerantz, Joshua Cohen, Yonatan Belinkov, Yuval Globerson, Yuval Levy, and Yoav Shoham. Jamba: Hybrid transformer–mamba language models. In *International Conference on Learning Representations*, volume 2025, pp. 67959–67984, 2025. URL [https://proceedings.iclr.cc/paper\\_files/paper/2025/hash/a9ed43fa31dc8b4a7d7a673d713dcb5f-Abstract-Conference.html](https://proceedings.iclr.cc/paper_files/paper/2025/hash/a9ed43fa31dc8b4a7d7a673d713dcb5f-Abstract-Conference.html).
- Yucheng Li, Huiqiang Jiang, Qianhui Wu, Xufang Luo, Surin Ahn, Chengruidong Zhang, Amir Abdi, Dongsheng Li, Jianfeng Gao, Yuqing Yang, and Lili Qiu. SCBench: A KV cache-centric analysis of long-context methods. In *International*

- Conference on Learning Representations*, volume 2025, pp. 66063–66093, 2025. URL [https://proceedings.iclr.cc/paper\\_files/paper/2025/hash/a540b17fb2295c736d5afd6c507acf66-Abstract-Conference.html](https://proceedings.iclr.cc/paper_files/paper/2025/hash/a540b17fb2295c736d5afd6c507acf66-Abstract-Conference.html).
- Hanzuo Liu, Xuan Qi, Chunyu Liu, Haotian Zhong, Yulong Wang, Key, Rayying, Alex Lamb, and Mingyu Gao. CoMem: Reusing transformer depth across queries with persistent intermediate residuals. In *COLM 2026 Workshop on Efficient Reasoning*, 2026a.
- Hanzuo Liu, Xuan Qi, Chunyu Liu, Haotian Zhong, Yulong Wang, Rayying, Key, Alex Lamb, and Mingyu Gao. Understanding is done early: A depth division of labor in large language models and its use for unbounded-context memory. *arXiv preprint arXiv:2607.28263*, 2026b. doi: 10.48550/arXiv.2607.28263. URL <https://arxiv.org/abs/2607.28263>.
- Yifei Liu, Juntong Wu, Yang Liu, Junhao Hu, Minghao Li, Xiaoxu Chen, and Weihang Chen. HYPIC: Accelerating hybrid-attention LLM serving with position-independent caching. *arXiv preprint arXiv:2607.01299*, 2026c. doi: 10.48550/arXiv.2607.01299. URL <https://arxiv.org/abs/2607.01299>.
- Rui Pan, Zhuang Wang, Zhen Jia, Can Karakus, Luca Zancato, Tri Dao, Yida Wang, and Ravi Netravali. Marconi: Prefix caching for the era of hybrid LLMs. In *Proceedings of Machine Learning and Systems*, volume 7. MLSys, 2025. URL [https://proceedings.mlsys.org/paper\\_files/paper/2025/hash/7c180af017258d239bac6248d1eb26ac-Abstract-Conference.html](https://proceedings.mlsys.org/paper_files/paper/2025/hash/7c180af017258d239bac6248d1eb26ac-Abstract-Conference.html).
- Hung Viet Pham, Thibaud Lutellier, Weizhen Qi, and Lin Tan. CRADLE: Cross-backend validation to detect and localize bugs in deep learning libraries. In *2019 IEEE/ACM 41st International Conference on Software Engineering (ICSE)*, pp. 1027–1038. IEEE, 2019. doi: 10.1109/ICSE.2019.00107. URL <https://doi.org/10.1109/ICSE.2019.00107>.
- Qwen Team. Qwen3.5: Towards native multimodal agents. Official Qwen technical blog, February 2026a. URL <https://qwen.ai/blog?id=qwen3.5>.
- Qwen Team. Qwen3.5-35B-A3B-Base model card. Hugging Face model repository, 2026b. URL <https://huggingface.co/Qwen/Qwen3.5-35B-A3B-Base/blob/main/README.md>. Accessed 2026-08-16.
- Jack W. Rae, Anna Potapenko, Siddhant M. Jayakumar, Chloe Hillier, and Timothy P. Lillicrap. Compressive transformers for long-range sequence modelling. In *International Conference on Learning Representations*, 2020. URL [https://iclr.cc/virtual\\_2020/poster\\_SylKikSYDH.html](https://iclr.cc/virtual_2020/poster_SylKikSYDH.html).
- Vikranth Srivatsa, Zijian He, Reyna Abhyankar, Dongming Li, and Yiyang Zhang. Preble: Efficient distributed prompt scheduling for LLM serving. In *International Conference on Learning Representations*, volume 2025, pp. 37057–37082, 2025. URL [https://proceedings.iclr.cc/paper\\_files/paper/2025/hash/5bc342f48de8264779952fac378f96dc-Abstract-Conference.html](https://proceedings.iclr.cc/paper_files/paper/2025/hash/5bc342f48de8264779952fac378f96dc-Abstract-Conference.html).
- Zihao Wang, Bin Cui, and Shaoduo Gan. SqueezeAttention: 2d management of KV-cache in LLM inference via layer-wise optimal budget. In *International Conference on Learning Representations*, volume 2025, pp. 23593–23606, 2025. URL [https://proceedings.iclr.cc/paper\\_files/paper/2025/hash/3b0a8df568ec496a717566a7f8158aaa-Abstract-Conference.html](https://proceedings.iclr.cc/paper_files/paper/2025/hash/3b0a8df568ec496a717566a7f8158aaa-Abstract-Conference.html).
- Dongwei Xiao, Zhibo Liu, Yuanyuan Yuan, Qi Pang, and Shuai Wang. Metamorphic testing of deep learning compilers. *Proceedings of the ACM on Measurement and Analysis of Computing Systems*, 6(1):1–28, 2022. doi: 10.1145/3508035. URL <https://doi.org/10.1145/3508035>.
- Songlin Yang, Jan Kautz, and Ali Hatamizadeh. Gated delta networks: Improving Mamba2 with delta rule. In *International Conference on Learning Representations*, volume 2025, pp. 29687–29707, 2025. URL [https://proceedings.iclr.cc/paper\\_files/paper/2025/hash/4904fad153f6434a7bcf04465d4be2cc-Abstract-Conference.html](https://proceedings.iclr.cc/paper_files/paper/2025/hash/4904fad153f6434a7bcf04465d4be2cc-Abstract-Conference.html).

Lu Ye, Ze Tao, Yong Huang, and Yang Li. ChunkAttention: Efficient self-attention with prefix-aware KV cache and two-phase partition. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 11608–11620, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.623. URL <https://aclanthology.org/2024.acl-long.623/>.

Lianmin Zheng, Liangsheng Yin, Zhiqiang Xie, Chuyue Sun, Jeff Huang, Cody Hao Yu, Shiyi Cao, Christos Kozyrakis, Ion Stoica, Joseph E. Gonzalez, Clark Barrett, and Ying Sheng. SGLang: Efficient execution of structured language model programs. In *Advances in Neural Information Processing Systems*, volume 37, pp. 62557–62583. Curran Associates, Inc., 2024. doi: 10.52202/079017-2000. URL [https://proceedings.neurips.cc/paper\\_files/paper/2024/hash/724be4472168f31ba1c9ac630f15dec8-Abstract-Conference.html](https://proceedings.neurips.cc/paper_files/paper/2024/hash/724be4472168f31ba1c9ac630f15dec8-Abstract-Conference.html).

## A ETHICS STATEMENT

This work introduces no human-subject data collection and evaluates only systems behavior on the PG-19 training split. It does not assess model safety, bias, privacy, or downstream deployment quality. Memory efficiency can lower deployment cost but may also increase the scale of model use; those broader effects are outside this controlled systems study.

## B COMPARISON LATTICE AND SAMPLING UNITS

This appendix describes the earlier capacity cohort only; its counts are not pooled with the pre-specified-window primary factorial.

|                                |               |               |               |                |                |                |
|--------------------------------|---------------|---------------|---------------|----------------|----------------|----------------|
| Token trajectories             | 8<br>pass     | 16<br>pass    | 32<br>pass    | 64<br>pass     | 128<br>pass    | 256<br>pass    |
| Paired full-vocabulary logits  | 64<br>pass    | 128<br>pass   | 256<br>pass   | 512<br>pass    | 1,024<br>pass  | 2,048<br>pass  |
| Logical KV (10 layers)         | 80<br>pass    | 160<br>pass   | 320<br>pass   | 640<br>pass    | 1,280<br>pass  | 2,560<br>pass  |
| GDN state (60 tensors/request) | 8<br>pass     | 16<br>pass    | 32<br>pass    | 64<br>pass     | 128<br>pass    | 256<br>pass    |
| Attention callable contract    | 1,280<br>pass | 2,560<br>pass | 5,120<br>pass | 10,240<br>pass | 20,480<br>pass | 40,960<br>pass |
|                                | $N = 1$       | $N = 2$       | $N = 4$       | $N = 8$        | $N = 16$       | $N = 32$       |

Resident fan-out (eight independent PG-19 books/ranks per column)

Totals: 504 request pairs; 4,032 paired logit comparisons; 5,040 KV layer digests; 504 GDN digests; 80,640 attention-call records. All compared artifacts passed.

Figure 4: Deterministic comparison counts by resident fan-out. Eight books are independent workload sources, each with one 32-query bank. Query sets are nested prefixes:  $N = 1$  reuses query 0,  $N = 2$  queries 0–1, and so on. Consequently, 504 paired request instances reduce to 256 unique queries, and only request indices 0–15 on each rank occur at two or more fan-outs. The cells count derived comparisons, not stochastic replicates.

Within this earlier two-arm cohort, all 504 paired request instances also match on the registered token, full-logit, logical-KV, and GDN records, and the 80,640 archived attention-call receipts use one fixed Python callable without a recorded path violation. These archival controls do not replace the primary four-cell RR2 evidence in Section 5. The nested fan-outs cover 256 unique queries rather than 504 independent examples: 128 queries recur at a larger fan-out and yield 248 non-vacuous cross- $N$  comparisons, all matching. Across the two arms, the ledger contains 4,032 paired full-vocabulary step comparisons and 8,064 arm-specific model steps. Full logits and states are not archived directly; replay instead checks shape/dtype-bound canonical CPU-FP32 logit bytes and logical-KV/GDN digests.

## C PROTOCOL GEOMETRY

Table 3: Frozen primary-factorial execution geometry. The eight ranks provide distinct books and query banks, not repeated stochastic measurements.

| Component        | Frozen value                                               |
|------------------|------------------------------------------------------------|
| Model            | Qwen3.5-35B-A3B; 10 attention + 30 GDN layers              |
| Hardware         | $8 \times$ H20-3e; one book/rank                           |
| KV backend       | vLLM 0.26 fused attention; BF16/Q16; page 128              |
| Document / query | 4,095 / 32 tokens; PG-19 train only                        |
| Generation       | 8 argmax tokens; 39 tokens appended to state               |
| Resident counts  | 1, 8, 32; four KV-by-GDN ownership cells                   |
| Schedule         | all requests live; sequential round-major, one CUDA stream |
| Validation       | 96 factor configurations; separate memory/witness cells    |
| Fault campaign   | $N = 2$ ; nine live mutants with rebuilt matched controls  |

Table 4: Cohort-to-claim authorization. Cohorts are never pooled; each row supports only the listed inference.

| Cohort                         | Frozen scope                                                                                                                                               | Authorized claim / explicit exclusion                                                                                                                                                                                                                                 |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary factorial (RR2)        | 8 books; $N = 1, 8, 32$ ; four KV $\times$ GDN cells; 8 decode steps                                                                                       | ownership lifecycle, cross-cell equivalence, selected attention oracle, live-mutant sensitivity, and four-cell allocator endpoints; not transfer or scheduler evidence                                                                                                |
| Earlier capacity               | 8 books; $N = 1, 2, 4, 8, 16, 32$ ; two KV arms                                                                                                            | cross- $N$ comparison counts and source-retaining capacity denominators only; not a primary-factorial replicate                                                                                                                                                       |
| Fresh-process control          | one independent $N = 2$ process; one clean path and M1                                                                                                     | input reconstruction, cleanup, and one intended-gate reproduction; not pooled with the factorial                                                                                                                                                                      |
| Lifecycle transfer             | 8 books; aligned 4,096-token prefix; $N = 4$ ; two cancelled slots                                                                                         | exact outputs/state, zero-scrub, exact reuse, and stale-lease rejection on the same adapter; not scheduler, concurrency, or performance evidence                                                                                                                      |
| Scheduler interleave           | 8 books; $N = 4$ ; page/prefix/tail geometries 64/3,985/17 and 128/4,033/65; cancel slot 3 after round 2                                                   | deterministic request-step interleaving, zero-scrub, epoch change, exact reuse, uninterrupted-control equality, and three frozen scheduler-fault gates; not concurrent kernels, continuous-batching performance, detection rate, or production end-to-end correctness |
| GDN transition oracle          | 1 frozen book/window; layers 0, 10, 20, 38; 4 query tokens                                                                                                 | clean recurrence outputs/states and four seeded wrong-transition rejections conditional on post-normalization inputs; not all GDN layers or end-to-end semantics                                                                                                      |
| M8/M9 detector debug           | M8 and M9 plus separate matched controls; intended gate suppressed on mutant path                                                                          | M9 reaches an ordinary paired-view runtime assertion; M8 reaches only the injected sentinel; no complete detector matrix or rate                                                                                                                                      |
| Mac common-mode control        | Apple M4 Pro; six documents/queries; five fresh MLX processes; vanilla dense plus CoMem BF16/Q8/Q4 at depth 7                                              | separates split-family agreement from agreement with vanilla dense; motivation only and separate from the H20 case                                                                                                                                                    |
| H20 deployment context         | 8 LongBench validation workloads; vanilla dense, full-prefix KV reuse, and CoMem Q8/Q4/mixed variants (six configurations total) with 3 timing repeats     | same-stack reusable-state, TTFT, TPOT, tokens/s, and mean F1 comparison; not pooled with RR2 and not scheduler/continuous-batching evidence                                                                                                                           |
| Matched serving controls       | 8 LongBench validation workloads; vLLM 0.26 and SGLang 0.5.17 cache off/on plus 24 fixed-configuration HYPIC timing/quality cells; one H20-3e per workload | within-block cache-hit, quality, and client-wall timing context; not pooled with in-process CoMem timings, not a cross-framework leaderboard, and not continuous-batching evidence                                                                                    |
| HYPIC retained-state receipts  | 8 Prefix Cache + 8 HYPIC cells on the same fixed TP=1 model/slice; receipt-instrumented SGLang lifecycle                                                   | median target-entry-owned physical tensor payload only; not timing, NVML/process memory, capacity, continuous batching, or ForkAudit ownership evidence                                                                                                               |
| Related-work operator transfer | Hydragen at $N = 8/32$ on one captured layer-3 attention family; Palu on one layer with 8 calibration and 1 held-out PG19 rows                             | same-checkpoint numerical and logical-state trade-offs only; not an all-layer implementation, LongBench/quality result, speedup claim, or jointly optimized system                                                                                                    |

Table 5: Decision status for the seven contract targets in the frozen case. Equality denotes canonical byte-digest identity as recorded. Overall status remains partial because dispatch lacks a per-call compiled-binary and autotuning-choice binding. “RC” means receipt-complete only within trusted producer capture and the target boundary in Section 3; the frozen-identity row covers one terminally closed formal run, not independent producer recapture or comprehensive system correctness.

| Target                 | Status  | Decisive witness                                                                                                      | Boundary                                                          |
|------------------------|---------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Frozen identity        | RC      | identical preflight/terminal source ledgers, terminal model rehash, and stable data/query/protocol/execution bindings | one validated formal run; not independent producer recapture      |
| Prefix immutability    | RC      | physical document-KV digests and GDN base guards across all three phases                                              | one frozen configuration/geometry                                 |
| Private ownership      | RC      | all-pairs request–base and request–peer range replay after transition                                                 | pointer-free conservative intervals                               |
| Tail-safe append       | RC      | one-time 127-token private-tail copy before append                                                                    | one partial-tail geometry                                         |
| Dispatch provenance    | partial | Python callable, query/position/mask/append/scale receipts; four compiled cache artifacts bound                       | no per-call compiled-binary or explicit autotuning-choice binding |
| Cross-arm equivalence  | RC      | tokens, shape/dtype-bound CPU-FP32 logit digests, logical KV, and GDN                                                 | one sequential schedule                                           |
| Cross- $N$ consistency | RC      | 288 all-arm adjacent-fan-out comparisons over 72 rank–query identities                                                | relational, not an independent oracle                             |

Selected attention and GDN numerical oracles, the nine targeted live faults, and the aligned lifecycle cohort are supporting checks rather than additional contract targets. Their selected captured-input and same-adaptor boundaries are reported separately in Sections 4–5 and Appendix Sections F–H.

## D POINTER-FREE STORAGE WITNESS SCHEMA

**Target-to-record dependencies.** The following matrix is normative. I, L, E, and F denote the Identity, Lifecycle, Execution, and Target/fault classes in Table 15; parenthesized entries name mandatory fields. Missing any marked field makes that target open. A preregistered geometry may make tail safety not applicable; the absence of a compiled-binary fingerprint leaves dispatch partial rather than silently full. A missing F field invalidates the corresponding fault row, but does not change an otherwise independent seven-target witness.

| Target                         | I                                      | L                                       | E                                                  | F        | Blind predicate                             |
|--------------------------------|----------------------------------------|-----------------------------------------|----------------------------------------------------|----------|---------------------------------------------|
| Frozen identity                | code, model, data, protocol; role map  | —                                       | —                                                  | N/A      | exact bytes and complete roles              |
| Prefix immutability            | document/base start digests            | roles; setup and final guards           | canonical content digests                          | N/A      | start = final                               |
| Private ownership              | request/base/private roles             | setup, first-transition, final bindings | normalized storage ranges                          | N/A      | non-vacuous all-pairs disjointness          |
| Tail-safe append               | page geometry; document/private roles  | first append event                      | ordered append and copied-tail receipt             | N/A      | one private copy before write               |
| Dispatch provenance            | callable and protocol identity         | —                                       | ordered call, query, mask, position, append, scale | N/A      | schedule and callable equality; no fallback |
| Cross-arm equivalence          | factor, query, and schedule identities | final bindings                          | token, logit, logical-KV, GDN records              | N/A      | canonical semantic equality                 |
| Cross- $N$ consistency         | query and fan-out identities           | —                                       | the same canonical semantic records                | N/A      | smallest- $N$ equals each larger $N$        |
| Fault sensitivity (supporting) | frozen target/gate map                 | matched-control rebuild and cleanup     | injected outcome and failed predicates             | required | target, injection, expected gate, outcome   |

Each tensor-view row has the exact fields `storage_id`, `shape`, `stride`, `storage_offset`, `dtype`, `device`, `storage_nbytes`, `tensor_nbytes`, and `content_sha256`; absolute addresses are forbidden. Within each phase, the producer assigns `storage-0000`, `storage-0001`, and so on in first-observation order to backing storages. Reusing an ID with a different device or storage size is invalid. Phase capture IDs are unique, whereas request and persistent lifecycle-guard IDs must remain stable across setup, first transition, and final generation.

Let element size be  $e$ , storage offset  $o$ , shape  $s_i$ , and stride  $t_i$ . The independent replay initializes  $m = M = o$  and, for every dimension, computes

$$d_i = (s_i - 1)t_i, \quad m += \min(d_i, 0), \quad M += \max(d_i, 0). \quad (2)$$

The conservative half-open byte interval is

$$I = [em, e(M + 1)), \quad (3)$$

which must be non-negative and contained in `storage_nbytes`; also `tensor_nbytes` =  $e \prod_i s_i$ . Two rows overlap iff their normalized storage IDs match and  $I_a^{\text{start}} < I_b^{\text{end}}$  and  $I_b^{\text{start}} < I_a^{\text{end}}$ . Exact aliasing additionally requires equality of interval, geometry, dtype/device, storage and tensor sizes, and content digest. This bounding interval can conservatively flag an exotic interleaved view, but it cannot miss a shared byte.

At setup, every recurrent-state coordinate must either exactly alias the persistent state (borrowed-base arm) or be disjoint from it (materialized arm). After a request’s first transition, its 60 mutable rows must be disjoint from the base and every peer; incomplete borrowed requests must remain exact aliases. At the final phase all requests must be disjoint. Independent binding-token receipts additionally require unchanged tokens for incomplete requests and changed binding/storage tokens for completed requests. The package applies this schema to all 96 timelines and 288 phase artifacts; nine adversarial tests cover exact aliasing, partial overlap, adjacent intervals, negative strides, conflicting ID reuse, pointer injection, capture/guard reuse, and cross-coordinate request–base and request–peer aliasing.

## E COMPLETE CAPACITY LEDGER

Table 6: Memory scaling under three explicitly named estimands. The controlled pool retains the common source in both arms and is therefore a controlled contrast, not an optimized production full-copy baseline. Post-pack and full recorded-lifecycle PyTorch peaks are allocator counters, not NVML usage or service capacity.

| $N$ | Controlled pool (MiB) |        | Post-pack prod. peak (GiB) |        | Full lifecycle peak (GiB) |        |
|-----|-----------------------|--------|----------------------------|--------|---------------------------|--------|
|     | Materialize           | Shared | Materialize                | Shared | Materialize               | Shared |
| 1   | 170                   | 85     | 64.896                     | 64.813 | 67.434                    | 67.434 |
| 2   | 260                   | 90     | 65.048                     | 64.881 | 67.434                    | 67.434 |
| 4   | 440                   | 100    | 65.345                     | 65.011 | 67.434                    | 67.434 |
| 8   | 800                   | 120    | 65.936                     | 65.270 | 67.434                    | 67.434 |
| 16  | 1520                  | 160    | 67.126                     | 65.792 | 67.434                    | 67.434 |
| 32  | 2960                  | 240    | 69.498                     | 66.837 | 69.498                    | 67.434 |

The source-retaining analytic pool is exactly  $80 + 90N$  MiB for Source+Materialize and  $80 + 5N$  MiB for Shared-Documents. The post-pack production peak is the maximum of resident setup and generation after the phase reset. The full recorded-lifecycle peak is the maximum over prefill, Q16 pack, resident setup, and generation. Values were identical across all eight ranks; this deterministic run does not support a stochastic confidence interval.

These values are source-retaining controls: the deterministic pools are  $80 + 90N$  MiB for materialization and  $80 + 5N$  MiB for sharing. Removing the retained materialized source is only an arithmetic counterfactual,  $80(N - 1)$  MiB; at  $N = 32$  it gives a 2,480 MiB difference and has no allocator measurement. For the executed source-retaining cohort, the post-pack production peaks differ by 2,857,268,224 bytes (3.8289%; ratio 1.0398 $\times$ ), while the full recorded-lifecycle peaks differ by a 2,216,007,168-byte amount (2.9696%; ratio 1.0306 $\times$ ). These phase-bounded peaks are PyTorch allocated counters, not NVML memory or admitted service capacity.

## F EXECUTED FAULT ROWS AND INVARIANT MAP

The row-level campaign is kept out of the main paper because it supports diagnosis and audit coverage rather than a separate headline result. Separately, the CPU governance suite passes 17 tests and

rejects 28 explicit serialized-artifact tamper cases plus its registered helper/launcher faults. That result concerns schema and orchestration fail-closure only; it is not GPU semantic or performance evidence.

Table 7: Executed live faults. Every separately rebuilt matched-clean cell passed; every mutant was rejected at its gate fixed before execution and restored before disposal.

| ID | Injected live fault | Expected/observed gate          | Outcome  |
|----|---------------------|---------------------------------|----------|
| M1 | reservation alias   | KV_RESERVATION_DISJOINT         | detected |
| M2 | sequence swap       | KV_SEQUENCE_ID                  | detected |
| M3 | omit tail COW       | KV_TAIL_COW                     | detected |
| M4 | GDN-base alias      | gdn_completed_vs_base_disjoint  | detected |
| M5 | GDN peer alias      | gdn_completed_vs_peers_disjoint | detected |
| M6 | position off-by-one | POSITION_CANONICAL_VALUES       | detected |
| M7 | materialized mask   | MASK_CONTRACT                   | detected |
| M8 | wrong callable      | KERNEL_CALLABLE_ID              | detected |
| M9 | dense KV view       | KV_PAGED_VIEW                   | detected |

Table 8: Representative failures each target is designed to expose. These are the nine executed primary-factorial live mutants; the checks are not complete oracles.

| Target                | Intended sensitivity and observed boundary                                      |
|-----------------------|---------------------------------------------------------------------------------|
| Frozen identity       | wrong model/data/query bank, changed protocol, stale shard (schema adversaries) |
| Prefix immutability   | M3 omits partial-tail COW; physical document digest and ownership gates fail    |
| Private ownership     | M1 reservation alias, M4 request-base GDN alias, M5 request-peer GDN alias      |
| Sequence/view binding | M2 sequence swap and M9 dense-KV replacement                                    |
| Dispatch provenance   | M6 position offset, M7 materialized mask, M8 wrong callable                     |
| Factorial exactness   | behavior changes when either KV or GDN setup ownership changes                  |
| Independent oracle    | numerical error in a selected fused full-attention result                       |
| Raw replay            | internally inconsistent trajectory, storage, allocator, or aggregate fields     |

This executed set supports only a bounded diagnostic decomposition. A simple output-equality test has no pre-output observation for M1–M9. ForkAudit’s storage receipts localize M1, M4, and M5; lifecycle/append receipts localize M2 and M3; and call-contract receipts localize M6–M9 before semantic output. This comparison identifies which recorded predicate adds diagnosis for these positive controls; it is not a measured detector rate, held-out-fault study, or claim that conventional tests could not be written to catch the same faults.

## G COMPONENT-BY-COMPONENT NOVELTY BOUNDARY

Table 9: Primary-source positioning. “Evidence focus” describes what the cited work evaluates; it is not a claim that the work lacks all other checks.

| Work                  | Primary mechanism                                                                                           | Evidence focus / relation to FORKAUDIT                                                                                                                                                 |
|-----------------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PagedAttention        | Paged KV allocation and sharing (Kwon et al., 2023)                                                         | Serving throughput and memory utilization; supplies the block-sharing substrate.                                                                                                       |
| SGLang / Prompt Cache | Prefix reuse in structured or modular prompts (Zheng et al., 2024; Gim et al., 2024)                        | Runtime performance and output behavior; establish reuse and positional handling as prior art.                                                                                         |
| CoMem                 | One persistent residual per token at split $j$ ; bounded selection; suffix replay (Liu et al., 2026a;b)     | Motivates the reusable document object; the bounded deployment table supplies same-stack context, while the contribution here remains the hybrid-state ownership audit.                |
| Marconi               | Admission and eviction for hybrid-model prefixes (Pan et al., 2025)                                         | Hybrid cache hit rate and time to first token; closest state-heterogeneous caching system, but a different policy-level contribution.                                                  |
| SCBench               | Lifecycle-oriented KV-cache benchmark (Li et al., 2025)                                                     | Breadth over tasks and cache phases; exposes breadth absent from our single reference case.                                                                                            |
| MT-DLComp / CRADLE    | Metamorphic compiler testing and cross-backend differential checking (Xiao et al., 2022; Pham et al., 2019) | Direct testing precedents; our audit adds ownership witnesses and a layerwise dense oracle, but not an independent end-to-end backend.                                                 |
| DNNMem                | Component-wise GPU-memory accounting (Gao et al., 2020)                                                     | Supports denominator separation; it is an estimator, not a request-ownership audit.                                                                                                    |
| FORKAUDIT             | Ownership/call/accounting ledger                                                                            | One mutation-tested KV-by-GDN ownership factorial, a bounded FP32 attention oracle, and same-adapter lifecycle/scheduler-contract checks; no new cache policy or production scheduler. |

Table 10: Closest established precedent for each audit obligation. The final column states the narrower integration contributed by FORKAUDIT; it does not claim that any individual mechanism is new.

| Audit obligation                   | Closest established precedent                                                                                         | Integrated obligation in this work                                                                                                                      |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Frozen identity                    | Controlled variants in metamorphic and differential testing (Chen et al., 2018; Xiao et al., 2022; Pham et al., 2019) | Bind model, data, query bank, callable contract, and protocol bytes to every ownership receipt.                                                         |
| Document immutability and tail COW | PagedAttention block sharing and copy-on-write for shared prefixes (Kwon et al., 2023)                                | Replay physical document pages, padding, active tables, and the partial-tail detach event.                                                              |
| Recurrent-state ownership          | Hybrid recurrent states and their in-place update constraint (Yang et al., 2025; Pan et al., 2025)                    | Witness request-base and request-peer ranges before and after each request’s first transition.                                                          |
| Dispatch provenance                | Cross-backend differential checking exposes implementation-dependent behavior (Pham et al., 2019)                     | Hold the backend fixed and record callable, position values, mask, scale, and append history per call.                                                  |
| Cross-arm and cross- $N$ relations | Metamorphic relations over semantics-preserving transformations (Chen et al., 2018; Xiao et al., 2022)                | Treat ownership policy and resident fan-out as transformations that must preserve registered request observables.                                       |
| Bounded numerical oracle           | Independent backend comparison (Pham et al., 2019)                                                                    | Reconstruct selected attention inputs for dense IEEE-FP32 math and selected post-normalization GDN inputs for a candidate-import-free NumPy recurrence. |
| Memory denominators                | Component-wise GPU-memory accounting (Gao et al., 2020)                                                               | Keep page geometry, allocator current/peak, and unmeasured process/service capacity as separate claim domains.                                          |

## H SUPPORTING CONTROL EXPERIMENTS

**Published systems context.** Table 11 reports each cited system only in its native protocol and keeps all non-commensurate metrics outside our matched panels. This unpooled table is literature context, not ForkAudit evidence.

Table 11: Unpooled published-systems context in each paper’s native protocol. These values show the scale and evaluation breadth of related systems, *not* a cross-paper leaderboard or ForkAudit evidence: models, hardware, batching, baselines, and metrics differ. Consequently, no row is merged with our same-stack H20 Table 2, and the absence of a common score must not be read as a negative result.

| System                                       | Native method and study setting                                                                                                                                               | Reported efficiency in its native protocol                                                                       | Quality and comparability boundary                                                                         |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| PagedAttention / vLLM (Kwon et al., 2023)    | Paged KV allocation plus serving engine<br><i>Setting</i> : LLaMA-7B on A10G and LLaMA-13B on A100-40GB; vLLM serving engine                                                  | 2–4 $\times$ serving throughput at similar latency vs. FasterTransformer/Orca                                    | No common LongBench F1; exact cache-management method                                                      |
| Prompt Cache (Gim et al., 2024)              | Modular attention-state reuse<br><i>Setting</i> : Llama2/MPT/Falcon; RTX 4090, A40, and A100; HuggingFace-style prototype                                                     | 1.5–3.1 $\times$ TTFT (CPU-resident modules); 3.7–11.7 $\times$ (GPU-resident) on its 12-dataset LongBench suite | LongBench accuracy evaluated, but with different models and hardware                                       |
| SGLang / RadixAttention (Zheng et al., 2024) | Prefix reuse plus structured-program runtime<br><i>Setting</i> : Llama-7B/Mixtral-8x7B and structured programs; A10G/A100; SGLang                                             | Up to 6.4 $\times$ throughput vs. prior inference systems                                                        | Application-suite result; no common LongBench F1                                                           |
| ChunkAttention (Ye et al., 2024)             | Prefix-aware KV chunks and attention kernel<br><i>Setting</i> : attention microbenchmarks; mainly A100-80GB, plus A100-40GB/RTX 4090; CUDA 11.8                               | 3.2–4.8 $\times$ attention-kernel speedup for 1K–4K shared prompts                                               | Kernel result; no end-to-end common task score                                                             |
| Hydragen (Juravsky et al., 2024)             | Exact batched shared-prefix attention<br><i>Setting</i> : CodeLlama-13B/34B shared-prefix batches; A100 (with H100/L40S microbenchmarks); custom HuggingFace runtime          | Up to 32 $\times$ CodeLlama-13B end-to-end throughput in large shared-prefix batches                             | Exact attention construction; different model and batching regime                                          |
| Palu (Chang et al., 2025)                    | Low-rank latent KV with reconstruction/fusion<br><i>Setting</i> : Llama/Mistral/LongChat quality studies; Llama-2-7B latency on one RTX 4090; HuggingFace plus custom kernels | 50% KV compression and up to 1.89 $\times$ RoPE-attention speedup; up to 2.91 $\times$ with quantization         | Llama/Mistral/LongChat native study; our bounded Qwen3.5 operator transfer is reported separately          |
| Preble (Srivatsa et al., 2025)               | Distributed prompt-sharing scheduling<br><i>Setting</i> : Mistral-7B/Llama-3-70B request traces; two–eight GPUs across A6000/H100 clusters; distributed serving               | 1.5–14.5 $\times$ average-latency and 2–10 $\times$ p99-latency improvement                                      | Distributed serving traces; no common task score                                                           |
| Marconi (Pan et al., 2025)                   | Admission/eviction for hybrid-model prefixes<br><i>Setting</i> : Attention–Mamba2 7B geometry; LMSys/ShareGPT/SWE-bench traces; CloudLab policy simulation                    | Up to 34.4 $\times$ token-hit rate and up to 71.1% or 617 ms lower TTFT                                          | Hybrid models and trace-driven policy study; no common LongBench F1                                        |
| HYPIC (Liu et al., 2026c)                    | Position-independent caching for hybrid full/linear-attention models<br><i>Setting</i> : Qwen3.5-35B-A3B main panels at TP=2 on 8 H20-3e; SGLang 0.5.14                       | 3.25 $\times$ lower TTFT on average and 1.66 $\times$ higher QPS over Prefix Cache                               | Four models/five workloads; our same-slice TP=1 timing and retained-state controls are reported separately |

**Same-checkpoint transfer detail.** Table 12 is the matched complement to the published context table: it uses our frozen Qwen3.5 checkpoint and H20-3e, binds the official Hydragen and Palu source revisions, and independently replays all raw operator sidecars. Because it does not execute all layers or LongBench, it is kept separate from Table 2.

Table 12: Bounded same-checkpoint H2O transfers. Hydragen uses frozen Qwen3.5 layer-3 post-RoPE attention; Palu uses eight 256-token PG19 calibration rows and one disjoint held-out row. Errors are relative  $\ell_2$  against the matched uncompressed operator; state ratios are logical selected-operator ratios, not process memory or end-to-end results.

| Port          | Setting       | Matched numerical result                     | Logical state result                    | Timing / boundary                |
|---------------|---------------|----------------------------------------------|-----------------------------------------|----------------------------------|
| Hydragen port | $N = 8$       | Hydragen-dense rel. $\ell_2 = 0.00255$       | 8.50 vs. 64.48 MiB (7.59× smaller)      | 0.287 vs. 0.0965 ms; no speedup  |
| Hydragen port | $N = 32$      | Hydragen-dense rel. $\ell_2 = 0.00262$       | 10.00 vs. 257.94 MiB (25.80× smaller)   | 0.290 vs. 0.259 ms; no speedup   |
| Palu port     | rank 64/head  | held-out K/V 0.487/0.549 (plain 0.711/0.839) | 512 vs. 2,048 B/token (4.00× smaller)   | projection only; no fused kernel |
| Palu port     | rank 128/head | held-out K/V 0.317/0.364 (plain 0.543/0.668) | 1,024 vs. 2,048 B/token (2.00× smaller) | projection only; no fused kernel |
| Palu port     | rank 192/head | held-out K/V 0.149/0.194 (plain 0.341/0.459) | 1,536 vs. 2,048 B/token (1.33× smaller) | projection only; no fused kernel |

Hydragen avoids prefix replication but has no speedup here. Palu improves over plain SVD, but residual error forbids an end-to-end quality claim. Neither is a jointly optimized combined system.

**Matched serving-framework and HYPIC context.** Table 13 gives separately timed vLLM and SGLang cache-off/on controls plus the official-code HYPIC Full Recompute, Prefix Cache, and full HYPIC arms under the shared Qwen3.5 streaming boundary. It remains a timing/quality-only panel and authorizes only within-framework interpretation; the separate 16-cell retained-state receipt cohort appears in Table 2.

Table 13: Unpooled timing/quality-only serving context on Qwen3.5 and the same eight-item Qasper/2WikiMQA slice. Each row uses one H2O-3e per item, a 4,096-token cap, greedy decoding, and at most 32 generated tokens. Times are OpenAI-compatible streaming client wall-clock; F1 is the eight-item mean. Hit is interpreted only against the disabled/full-recompute reference inside each framework block. No cross-framework speedup, retained-state, scheduler-throughput, capacity, or broad-quality claim is made.

| Framework     | Prefix reuse      | F1    | TTFT (s) | TPOT (ms) | tok/s | Hit |
|---------------|-------------------|-------|----------|-----------|-------|-----|
| vLLM 0.26     | off               | 39.63 | 1.445    | 56.31     | 4.70  | –   |
| vLLM 0.26     | align on          | 39.63 | 0.179    | 53.99     | 14.00 | 8/8 |
| SGLang 0.5.17 | off               | 39.14 | 0.283    | 53.38     | 12.73 | –   |
| SGLang 0.5.17 | Radix on          | 39.14 | 0.146    | 53.15     | 15.83 | 8/8 |
| HYPIC code    | full recompute    | 39.14 | 0.217    | 49.21     | 14.77 | –   |
| HYPIC code    | prefix cache      | 39.14 | 0.072    | 50.15     | 19.07 | 8/8 |
| HYPIC         | transition+seam 8 | 39.63 | 0.101    | 52.13     | 17.67 | 8/8 |

vLLM uses experimental Mamba align; SGLang 0.5.17 uses `extra_buffer` Radix caching. The HYPIC rows use the authors’ commit 98147c0 on SGLang 0.5.14, TP=1, full `transition.rope.recompute`, and an eight-token seam. Its full and prefix rows are controls from that same codebase; published TP=2 results and the separate retained-state receipt cohort are not inserted here.

**Hybrid-prefix policy context.** The separate preregistered Marconi trace extends the eight frozen workloads with a fixed synthetic follow-up turn and evaluates only prefix-policy token reuse under the artifact’s native Attention–Mamba2 geometry. It neither reruns LongBench quality nor supplies Qwen3.5 serving speed, so it is not pooled with the preceding panel or the primary audit.

Table 14: Unpooled preregistered policy-simulator context on a 128-event synthetic multi-turn extension of the eight frozen workloads. Values are token-hit percentages under the official Marconi artifact’s native Attention–Mamba2 geometry. The wrapper calls each policy’s own eviction routine after insertion to close actual tree size to the declared budget. These are not Qwen3.5 quality, latency, or throughput measurements and do not validate the primary ownership audit.

| Policy  | 5 GB         | 10 GB | 20 GB |
|---------|--------------|-------|-------|
| vLLM+   | 2.07         | 6.18  | 46.83 |
| SGLang+ | 90.78        | 93.86 | 93.86 |
| Marconi | <b>93.86</b> | 93.86 | 93.86 |

Table 15: Normative record classes. “Mandatory” means that absence makes the corresponding target open rather than silently not applicable.

| Record class | Mandatory payload                                                         | Independent replay                            |
|--------------|---------------------------------------------------------------------------|-----------------------------------------------|
| Identity     | frozen code, model, data, protocol, and object-to-role map                | exact bytes and role coverage                 |
| Lifecycle    | setup, first-transition, and final inventories and bindings               | normalized ranges, aliases, and disjointness  |
| Execution    | ordered call/append receipts, semantic records, named allocator snapshots | frozen schedule, digests, and phase equations |
| Target/fault | predeclared target, gate, matched control, injection, and outcome         | failed-predicate set and valid-negative class |

**Primary allocator endpoints.** Table 16 keeps the four ownership cells and their allocator denominators together under the primary 8-rank cohort.

Table 16: Primary  $8 \times H20-3e$  allocator results at  $N = 32$  (median across ranks; rank values coincide). Vanilla full-copy is the external control; the paged-prefix row is the closest same-stack prefix-sharing baseline. The remaining rows isolate recurrent-base ownership. All cells retain the persistent document source. Values are GiB above the frozen post-priming baseline; generation increment is the additional peak above allocation present at generation start.

| Empirical arm          | KV / GDN setup                        | Final | Peak  | Gen. incr. |
|------------------------|---------------------------------------|-------|-------|------------|
| Vanilla full-copy      | Full-copy KV / Materialized GDN base  | 4.901 | 4.920 | 0.019      |
| GDN ownership ablation | Full-copy KV / Borrowed GDN base      | 4.890 | 4.907 | 1.951      |
| Paged-prefix baseline  | Shared-doc KV / Materialized GDN base | 2.229 | 2.843 | 0.019      |
| Audited hybrid fork    | Shared-doc KV / Borrowed GDN base     | 2.229 | 2.843 | 1.950      |

**Aligned cancellation and reclamation lifecycle.** The separate lifecycle-transfer cohort uses eight ranks, eight distinct PG-19 training books, an aligned 4,096-token prefix, and  $N = 4$ . It is not pooled with the primary factorial.

Table 17: Lifecycle-transfer obligations on the existing Qwen3.5/vLLM Q16 adapter. Counts are deterministic rank outcomes, not stochastic replicates.

| Obligation                                                            | Result                          |
|-----------------------------------------------------------------------|---------------------------------|
| Frozen static/code/model/weight/raw bindings                          | pass; 14/14 weights, 8/8 shards |
| Full-vocabulary logits and generated tokens vs. uninterrupted control | exact, 8/8                      |
| Final logical KV and final GDN state                                  | exact, 8/8                      |
| Aligned prefix, immutable document, and zero partial-tail copy        | pass, 8/8                       |
| Cancelled-slot zero-scrub and exact reservation reassignment          | pass, 8/8                       |
| Stale epoch-0 rejection and matched epoch-1 acceptance                | 8/8; 0 escapes, 0 wrong gates   |

**Scheduler-managed request-step interleaving.** The separate formal extension keeps four requests resident, cancels slot 3 after its second round, zero-scrubs and increments its lease epoch, reassigns the exact reservation, and completes the replacement interleaved with the three survivors. It is not pooled with the primary factorial or the aligned lifecycle-transfer cohort.

Table 18: Formal scheduler-managed request-step interleaving on the frozen Qwen3.5/ForkAudit vLLM-Q16 stack. Clean counts are deterministic rank outcomes; fault counts are preregistered expected-gate outcomes, not a detection rate.

| Case             | Frozen geometry or intervention                                                            | Outcome                                                                    |
|------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Clean geometry 1 | page 64; 3,985-token prefix; 17-token document tail; 18-event cancel/replacement lifecycle | all semantic/storage gates and uninterrupted-control comparisons pass, 8/8 |
| Clean geometry 2 | page 128; 4,033-token prefix; 65-token document tail; same lifecycle                       | all semantic/storage gates and uninterrupted-control comparisons pass, 8/8 |
| FH1              | dispatch cancelled epoch-0 lease                                                           | STALE_SLOT_LEASE, 16/16                                                    |
| FH2              | dispatch lease under another request                                                       | LEASE_REQUEST_MISMATCH, 16/16                                              |
| FH3              | reclaim before zero scrub                                                                  | RECLAIM_NOT_ZERO, 16/16                                                    |

**Independent GDN transition oracle.** The oracle uses one frozen PG-19 window and four query-time layer transitions at layers 0, 10, 20, and 38. A single preregistration frozen before the successful run binds the post-native-q/k-normalization boundary, scale  $1/\sqrt{128}$ , tolerances, and seeded faults. Its NumPy reference starts at that captured boundary; no candidate module is imported by the reference.

Table 19: PDF-visible oracle selection and replay boundary. Every listed row was locked before its successful candidate execution; neither oracle validates upstream activation construction or end-to-end model behavior.

| Oracle    | Frozen rows / selection rule                                                                                                                                   | Captured boundary                                                   | Decision rule                                                                           | Replay availability                                                       |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Attention | rank $r$ uses layer $4r + 3$ and round $r$ , for $r \in \{0, \dots, 7\}$ ; request 0, $N = 1$ , shared-document KV + borrowed GDN; frozen rank/layer/round map | post-RoPE Q/K/V, positions, mask, scale, and ordered append shadows | dense CPU-FP32 rel. $L_2 \leq 0.005$                                                    | 8 raw rows and numerical sidecars; RR2 one-command replay                 |
| GDN       | layers 0, 10, 20, 38; four query tokens from one frozen PG-19 window                                                                                           | post-native-q/k-normalization recurrence; scale $1/\sqrt{128}$      | output rel. $L_2 \leq 0.005$ ; state rel. $L_2 \leq 10^{-4}$ ; seeded fault must reject | 4 clean + 4 fault decisions from 40 sidecars; separate one-command replay |

Table 20: Bounded GDN recurrence oracle. Relative errors compare native outputs and final states with the candidate-import-free NumPy FP32 recurrence.

| Layer | Output rel. $L_2$ | State rel. $L_2$      | Clean | Seeded fault       | Rejected |
|-------|-------------------|-----------------------|-------|--------------------|----------|
| 0     | 0.001584          | $1.23 \times 10^{-7}$ | yes   | omit decay         | yes      |
| 10    | 0.001652          | $1.36 \times 10^{-7}$ | yes   | complement $\beta$ | yes      |
| 20    | 0.001572          | $1.39 \times 10^{-7}$ | yes   | pre-decay memory   | yes      |
| 38    | 0.001638          | $1.30 \times 10^{-7}$ | yes   | roll value heads   | yes      |

**M8/M9 local detector debug.** The scoped debug run contains only M8, M9, and their matched-clean controls. M9 reaches the existing production paired-view assertion after its named gate is suppressed. M8 reaches an injected sentinel rather than an existing runtime assertion. Both mutant paths abort before output, so token/logit and cross-arm/cross- $N$  consequences are not observed. These debug rows are not formal evidence for a detector rate and are never combined with M1–M7.

**Single-request fair control.** The supporting run evaluates eight workloads with four materialize and four shared trials per workload under the same recorded attention callable. All full-logit trajectories match. Median per-workload shared/materialize ratios are 1.0090 for the cached-document request TTFT, 1.0133 for the complete continuation first model step, and 1.0006 for TPOT. Incremental peak allocated memory decreases, but reserved-memory peak does not. Because timings are short, fixed-scope, and not a deployment scheduler measurement, we draw no speed claim. The anonymous supplement includes this verified summary as context, not its raw trials; it is therefore not a primary replayable result of this paper package.

## I EXPANDED LIMITATIONS AND CLAIM BOUNDARY

The primary factorial covers one model, one H20 hardware family, one page size, 16-bit KV, and  $N \leq 32$ . It has one 4,095-token partial-tail geometry; the separate lifecycle cohort adds one aligned 4,096-token geometry and a fixed cancel/scrub/reclaim sequence on the same adapter. The formal scheduler extension adds two page/prefix/tail geometries and one deterministic request-step interleaving with cancellation and replacement, but not concurrent CUDA kernels. Requests are simultaneously resident but execute sequentially on one CUDA stream. The separate eight-workload deployment context evaluates CoMem Q8/Q4 state and single-request F1/timing, but the primary audit does not. No cohort evaluates continuous/ragged batching, eviction, arbitrary scheduler-level cancellation, multi-document serving, NVML memory, OOM capacity, or a broad downstream benchmark. Full-copy KV and materialized GDN bases are controlled ownership factors, not optimized production baselines. The GDN path is the Transformers Torch implementation, not an optimized recurrent kernel. The formal rerun hashes four compiled runtime-cache artifacts, but the callable ledger does not bind each call to a selected binary and exposes no explicit autotuning artifact.

The auxiliary serving panel uses one request stream per engine and compares each cache-enabled framework only with its own disabled reference. It does not measure continuous batching, eviction, admission control, maximum concurrency, or best-tuned performance. CUDA graphs were disabled in both pinned runs; the reported values describe that fixed configuration only.

The HYPIC timing/quality comparison uses the authors’ exact tracked commit on SGLang 0.5.14 in one fixed CUDA 12.9 / PyTorch 2.11 configuration, adapting Qwen3.5 execution from the paper’s TP=2 to TP=1 so every arm uses one H20-3e per workload. Cache-hit authority comes from OpenAI response usage. Each of its 24 timing/quality cells has one formal measurement after a discarded, prefix-disjoint warmup. A separate receipt-instrumented run uses the same model/slice and fixed TP=1 serving configuration to produce 8 Prefix Cache and 8 HYPIC retained-state cells; each completed cell binds raw output plus pre-measurement and terminal ownership receipts, and all 16 pass an external hash-bound replay. The instrumentation adds target-entry byte-range receipts and a lifecycle-idempotency overlay; Store therefore does not come from the unmodified timing run. Neither cohort establishes scheduler throughput, QPS, capacity, the paper’s TP=2 latency, or accuracy robustness beyond this eight-item slice.

The same-checkpoint related-work transfer is likewise operator-bounded: Hydragen uses one captured layer-3 shared-prefix attention family, and Palu uses one layer with eight calibration rows and one held-out row. Neither executes all model layers, a fused Palu kernel, LongBench, or a jointly optimized Hydragen/Palu-plus-CoMem serving path.

The page equations are deterministic identities, not learned laws; there is one observation per rank-configuration cell and no stochastic replication. Most runtime tensors are represented by hashes and equality receipts rather than archived in full. The attention and GDN oracle rows are the exception and ship numerical sidecars, but cover only selected captured-input operator boundaries rather than an end-to-end model. The fully preregistered GDN run starts after native q/k normalization and therefore does not validate that upstream boundary. Pointer-free normalized intervals plus raw-byte hashes make the captured storage evidence replayable, not a proof against a malicious producer before capture. The nine mutants are targeted positive controls; the M8/M9 debug adds one ordinary downstream catch and one unmeasured runtime-detector cell, not incremental value. The scheduler extension’s three frozen faults are expected-gate outcomes, not a detection rate or evidence of fault-set completeness. Broader systems claims would additionally require a production serving scheduler with concurrent kernels, optimized controls, alignment/length sweeps, and process-level accounting.

## J ARTIFACT AND CLAIM MAP

Table 21: Primary artifact map. Relative paths are rooted at the accompanying research package. Metadata-redacted derivatives preserve numerical fields and state digests, while most full tensors still require the pinned model/data environment for independent numerical recomputation.

| Claim/evidence                                           | Machine-readable source                                                                                                                                                                           |
|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary RR2 replay package                               | evidence/round_04_rr2_package/                                                                                                                                                                    |
| Replay and package manifest                              | evidence/round_04_rr2_package/replay/run_replay.sh and<br>evidence/round_04_rr2_package/MANIFEST.json                                                                                             |
| Registered, derived, and raw evidence                    | upstream/forkaudit-summary.json, derived/derived_summary_v2.json, and upstream/raw/ under the RR2 package                                                                                         |
| Executed source and storage tests                        | executed_source/gpu/ and replay/test_storage_witness.py under the RR2 package                                                                                                                     |
| Identity, lifecycle, scheduler, and GDN extensions       | evidence/rr2_terminal_closure_a1_20260821/, evidence/lifecycle_transfer_reviewer_package/, evidence/round23_a2_scheduler_interleave/, and evidence/gdn_transition_oracle_preregistered_20260819d/ |
| Mac and H20 deployment controls                          | evidence/mac_m4_motivation/, evidence/h20_deployment_benchmark/, and their generators under scripts/                                                                                              |
| HYPIC retained-state receipts                            | evidence/related_work_same_protocol/hypic-retained-state-r34-trial1892234/                                                                                                                        |
| vLLM, SGLang, HYPIC timing/quality, and Marconi controls | evidence/related_work_same_protocol/; exact package members and one-command replays are indexed by the experiment registry                                                                        |
| Hydragen/Palu same-checkpoint transfers                  | evidence/round24_related_work_transfer/ and the corresponding sources under gpu/                                                                                                                  |
| Published context and table generator                    | literature/reported_system_context.json and scripts/generate_related_work_context_table.py                                                                                                        |
| Integrated bounded results and claim graph               | evidence/integrated_results.json, evidence/claim_evidence_map.tsv, and evidence/method_provenance.tsv                                                                                             |